Kyle Aaron Thompson

Auburn, Alabama | (256) 603-2486 | kat0028@auburn.edu | github.com/kat0028 | Active SECRET Clearance

Education

Auburn University — Auburn, Alabama

Ph.D. Mechanical Engineering, Minor in Aerospace Engineering — *Expected 2028*

M.S. Mechanical Engineering — *August 2025*

B.S. Mechanical Engineering, Magna Cum Laude — *August 2020*

- **Graduate GPA:** 4.0/4.0
- **Graduate Coursework:** Optimal Control of Aerospace Vehicles, Advanced Trajectory Optimization, Nonlinear Systems and Control, Adaptive Control, Advanced Discrete Control, Fundamentals of Navigation and Guidance, Fundamentals of GPS, Neural Networks, Missile Aerodynamics, Applied Mechatronics

Experience

Graduate Research Assistant – Auburn University — August 2020 – Present

GPS and Vehicle Dynamics Laboratory (GAVLAB) — *Auburn, Alabama*

- Conduct sponsored research in guidance, navigation, and control (GNC) of autonomous ground vehicles, focusing on guidance and control, with additional work on aerial platforms.
- Developed longitudinal and lateral guidance and control for Class 8 trucks, sedans, and racing platforms.
- Developed a GPS-independent leader–follower convoy system using UWB ranging and vehicle models, achieving 1–5 m path-duplication accuracy at 10–50 m following distances.
- Serve as controls, guidance, and software lead for Autonomous Tiger Racing (ATR), Auburn's Indy Autonomous Challenge (IAC) team.
- Developed the majority of the team's current software stack.
- Designed raceline trajectory optimizers formulated as nonlinear programs.
- Built real-time optimal feedback controllers as quadratic programs solved online with OSQP for reliable timing.
- Published multiple academic papers as first author and co-authored additional conference and journal papers.

United States Marine Corps Reserve — July 2014 – Present

GySgt 0869 – Platoon Sergeant

- Train and supervise an Operational Firing Platoon of approximately 35 Marines.
- Oversee safe operation of M142 HIMARS rocket artillery systems for Battery K, 2nd Battalion, 14th Marines.
- Developed and taught curriculum for the professional certification of HIMARS operators.
- Deployments:
  - Afghanistan 2019 – TFWS 19.1/2
  - Okinawa 2022–2023 – Sierra 3/12 UDP 23.1

Teledyne Brown Engineering, Inc. — April 2015 – July 2015; June 2016 – August 2016

Classified Lab Administrative Assistant — *Huntsville, Alabama*

- Secured the classified lab after hours, enforced security policies, and performed basic system administration.

Selected Publications

- **K. Thompson**, "Intelligent Emergency Stop System for Autonomous Ground Vehicles Using Minimum Jerk Trajectories and Path Tracking Linear Quadratic Regulator," *M.S. thesis*, Auburn University, Auburn, AL, Aug. 2025.
- **K. Thompson**, B. Jones, S. Martin, and D. Bevly, "GPS-Independent Autonomous Vehicle Convoying with UWB Ranging and Vehicle Models," *NDIA Ground Vehicle Systems Engineering and Technology Symposium (GVSETS)*, Novi, MI, Aug. 2022.

Skills

Guidance and Control: Classical control, LQR, MPC, trajectory optimization, path tracking, Kalman filtering
Languages: C++, Python, MATLAB; also C# (Windows), Swift (Apple), and Kotlin (Android)
Tools and Libraries: ROS/ROS 2, Qt, CasADi, IPOPT, OSQP, Git, LaTeX, and SolidWorks (3D CAD)
Computer and Software: Linux, Windows, macOS, Visual Studio, Xcode
Embedded Systems: Autonomy software on single-board computers and microcontrollers